



Barker
College

2023

YEAR 12

ASSESSMENT TASK 3

Mathematics Advanced

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Student Number

Staff Involved:

• DZB* • ARM • DXC • JAI • LAK

• LMD • ALY • ARP • PDJ • RJW

Wednesday 31st May, 8:30am

160 copies

General Instructions

- Working time – 55 minutes
- Write using blue or black pen
- Write your Barker Student number in the spaces indicated
- Write your answers in the spaces provided
- Board approved calculators may be used
- The HSC Reference Sheet is provided.
- Attempt all questions
- ALL necessary working must be shown
- Marks may not be awarded for careless or badly arranged work
- Diagrams are NOT to scale

Topics Covered

- Exponential and Logarithmic Functions
- Trigonometric Functions
- Motion and Rates

Section	Marks
Question 1	/ 12
Question 2	/ 10
Question 3	/ 11
Question 4	/ 13
Total Marks:	/ 46

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Question 1 (12 marks)

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Student number

(a) Differentiate with respect to x , leaving your answer in simplest form:

(α) $y = \ln 3x$ 1

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(β) $y = e^{(x^2)}$ 1

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(γ) $y = \frac{1}{2}\cos^2 x$ 2

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(δ) $y = \frac{\sin x}{e^{3x}}$ 3

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(b) Find the following indefinite integrals:

(α) $\int 2e^{2x+1} dx$ 1

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(β) $\int \frac{x^2}{x^3+1} dx$ 2

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(γ) $\int 4x \sec^2(x^2) dx$ 2

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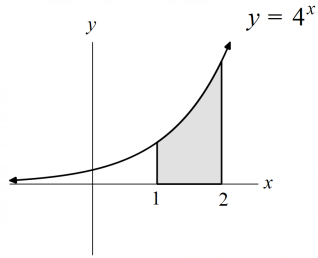
END OF QUESTION 1

Question 2 (10 marks)

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Student number

- (a) Evaluate the area under the curve $y = 4^x$, bounded by the x axis, $x = 1$ and $x = 2$. Round your answer to 2 decimal places. **2**



- (b) Find the equation of the **normal** to the curve $y = \sin 2x$ at the point on the curve where $x = \frac{\pi}{2}$. **4**

- (c) The height (h) of a cannonball, in metres, versus time (t), in seconds, is given by the equation $h = 4(t - 3)(10 - t)$.

- i) Calculate the time(s) when the cannonball is at a height of 24 metres. **2**

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- ii) Calculate the velocity the cannonball is travelling at 6 seconds. **2**

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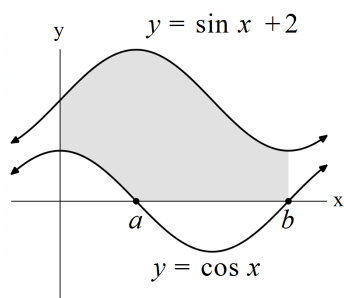
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END OF QUESTION 2

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(a) Consider the graph below.



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- [illegible]

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- This image shows a full page of primary-ruled paper. It features multiple horizontal rows, each consisting of two parallel dotted lines. These rows are designed to help young learners practice their handwriting by keeping letters at a consistent height and aligned correctly. The paper is otherwise blank, with no margins or additional markings.

8

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(a) An ocean buoy's displacement (x), in metres above sea level, is given by the equation $x = 0.8 \cos(\frac{\pi}{7}t)$, where t is measured in seconds.

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- [illegible]

- [illegible]

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- [illegible]

END OF ASSESSMENT

1a) α) $y = \ln 3x$
 $\frac{dy}{dx} = \frac{1}{3x} \times 3 = \frac{1}{x}$
 β) $y = e^{x^2}$
 $y' = e^{x^2} \times 2x$
 $y' = 2xe^{x^2}$
 γ) $y = \frac{1}{2} \cos^2 x = \frac{1}{2} (\cos x)^2$
 $y' = \frac{2}{2} (\cos x) \times -\sin x$
 $= -\sin x \cos x$
 δ) $y = \frac{\sin 2x}{e^{3x}}$ $u = \sin 2x$
 $y' = \frac{e^{3x} \cos 2x - 3e^{3x} \sin x}{(e^{3x})^2}$ $u' = \cos 2x$
 $= \frac{\cos x - 3 \sin x}{e^{3x}}$ $v = e^{3x}$
 $v' = 3e^{3x}$
 b) $\int 2e^{2x+1} dx = \frac{2e^{2x+1}}{2} + C$
 $= e^{2x+1} + C$
 γ) $\int \frac{x^2}{x^3+1} dx$ $y = x^3+1$
 $= \frac{1}{3} \int \frac{3x^2}{x^3+1} dx$ $y' = 3x^2$
 $= \frac{1}{3} \ln |x^3+1| + C$
 δ) $\int 4x \sec^2(x^2) dx$
 $= 2 \int 2x \sec^2(x^2) dx$ $y = x^2$
 $= 2 \tan x^2 + C$ $y' = 2x$
 2a) $\int_1^2 4^x dx = \left[\frac{4^x}{\ln 4} \right]_1^2$
 $= \frac{16}{\ln 4} - \frac{4}{\ln 4} = \frac{12}{\ln 4}$
 $= 8.65617 \dots$
 $= 8.66$ (2 dec. places)

2b) $y = \sin 2x$
 $y' = 2 \cos 2x$
 When $x = \frac{\pi}{2}$
 $y' = 2 \cos \pi = 2 \times -1 = -2$
 $\therefore \text{slope} = -2$
 $y = \sin 2x \times \frac{\pi}{2} = \sin \pi = 0$
 $y - 0 = \frac{1}{2} (x - \frac{\pi}{2})$
 $y = \frac{1}{2} x - \frac{\pi}{4}$
 $2x - 4y - \pi = 0$ (General form)
 c) i) $24 = 4(t-3)(10-t)$
 $6 = -t^2 + 13t - 30$
 $t^2 - 13t + 36 = 0$
 $(t-9)(t-4) = 0$
 $\therefore t = 4$ or $t = 9$ secs
 ii) $h = 4(-t^2 + 13t - 30)$
 $= -4t^2 + 52t - 120$
 $v = h' = -8t + 52$
 When $t = 6$
 $v = -8(6) + 52$
 $= 4$ m/s.
 3a) i) $a = \frac{\pi}{2}$, $b = \frac{3\pi}{2}$
 ii) $A = \int_0^{\frac{3\pi}{2}} \sin x + 2 dx - \int_0^{\frac{\pi}{2}} \cos x dx$
 $= [-\cos x + 2x]_0^{\frac{3\pi}{2}} - [\sin x]_0^{\frac{\pi}{2}}$
 $= ((-\cos \frac{3\pi}{2} + 3\pi) - (-\cos 0 + 0)) - (\sin \frac{\pi}{2} - \sin 0)$
 $= (0 + 3\pi + 1) - (1 - 0)$
 $= 3\pi + 1$
 b) $f(x) = x - x \ln x$
 $f'(x) = 1 - (\ln x + x \times \frac{1}{x})$
 $= 1 - \ln x - 1$
 $= -\ln x$
 ii) $\int -\ln(x^2) dx$
 $= 2 \int -\ln x dx + C$
 $= 2(x - x \ln x) + C$

3b) i) $f'(x) = -\ln x$
 possible max value when
 $f'(x) = 0$
 $0 = -\ln x$
 $0 = \ln x$
 $e^0 = x \therefore x = 1$

x	$\frac{1}{2}$	1	2
$f'(x)$	0.69	0	-0.69

 $\therefore \text{max @ } x = 1$
 When $x = 1$, $y = 1 - 1 \times \ln 1$
 $= 1$
 $\therefore \text{max value of } f(x) = 1$
 4a) i) when $t = 0$ $x = 0.8 \cos 0$
 $= 0.8$ m
 ii) $0.4 = 0.8 \cos \frac{\pi}{2} t$
 $\frac{1}{2} = \cos \frac{\pi}{2} t$
 $\frac{\pi}{2} t = \frac{\pi}{3} \therefore t = \frac{2}{3}$ secs
 b) i) when $t = 0$ $P = 0 + 200$
 $= 200$ million
 ii) $0 = e^{-0.01t} (5t + 200)$
 $5t + 200 = 0$
 $5t = -200$
 $t = -40$ hours
 iii) $P = 5te^{-0.01t} \times 0.01 + 5e^{-0.01t} - 2e^{-0.01t}$
 $= -0.05te^{-0.01t} + 3e^{-0.01t}$
 $= e^{-0.01t} (-0.05t + 3)$
 T.P. when $P' = 0$
 $0 = e^{-0.01t} (-0.05t + 3)$
 $-0.05t + 3 = 0$
 $-0.05t = -3$
 $t = 60$ hours.

4b) i) cont.

t	59	60	61
P'	0.028	0	-0.027

 $\therefore \text{max at } t = 60$
 When $t = 60$
 $P = 5 \times 60e^{-0.01 \times 60} + 200e^{-0.01 \times 60}$
 $= 274.405818$
 $= 274$ million (nearest million)